

Exam 1 foundation check

Yes or no

1 Shared lists

Two names refer to the same list. One name is used to append an item. Does the other name see that item?

Answer: Yes / No / Not sure. Confidence: _____

Reason in a few words: _____

2 Consecutive loops

Two loops run one after the other. Each visits n items and does constant work per item. Is their total running time proportional to n squared?

Answer: Yes / No / Not sure. Confidence: _____

Reason in a few words: _____

3 List indexing

For a nonempty Python list, must reading its last item scan all earlier items?

Answer: Yes / No / Not sure. Confidence: _____

Reason in a few words: _____

4 Sorting in place

A function rearranges the original list using constant extra space. Can it sort in place even if it returns None?

Answer: Yes / No / Not sure. Confidence: _____

Reason in a few words: _____

5 Rebinding a parameter

What prints? Choose one answer, or write "not sure".

```
def change(items):  
    items = items + [3]  
values = [1, 2]  
change(values)  
print(values)
```

- A. [1, 2]
- B. [1, 2, 3]
- C. [3]
- D. An error

Answer: _____ Confidence: _____

Multiple choice

Choose one answer per question, or write "not sure".

6 Nested loops

An outer loop runs n times. For each outer iteration, an inner loop runs n times and does constant work per iteration. What is its tightest worst-case Big-O cost?

- A. $O(1)$
- B. $O(n)$
- C. $O(n^2)$
- D. $O(\log n)$

Answer: _____ Confidence: _____

7 Dynamic array append

A dynamic array doubles its capacity when full. With n items already stored, what is the worst-case cost of one append?

- A. $O(1)$
- B. $O(n)$
- C. $O(n^2)$
- D. $O(\log n)$

Answer: _____ Confidence: _____

8 Size and capacity

A dynamic array stores 3 items in 8 reserved slots. What are its size and capacity?

- A. Size 8, capacity 3
- B. Size 3, capacity 3
- C. Size 8, capacity 8
- D. Size 3, capacity 8

Answer: _____ Confidence: _____

9 Stable sorting

Records (2, Ada), (1, Bo), (2, Cy) are sorted by number only. Which output preserves stability?

- A. (1, Bo), (2, Cy), (2, Ada)
- B. (1, Bo), (2, Ada), (2, Cy)
- C. (2, Ada), (1, Bo), (2, Cy)
- D. Either A or B

Answer: _____ Confidence: _____

10 Bubble sort

Trace this pseudocode on [4, 1, 3, 2]. What is the list after this single pass?

```
FOR each adjacent pair, from left to right
    IF the left value is greater than the right value
        SWAP the two values
```

- A. [1, 2, 3, 4]
- B. [4, 1, 3, 2]
- C. [1, 3, 2, 4]
- D. [4, 3, 2, 1]

Answer: _____ Confidence: _____

Short traces

Write the output and a short reason. You may write “not sure”.

11 Independent objects

```
class Counter:
    def __init__(self):
        self.value = 0
    def add(self):
        self.value += 1
a = Counter()
b = Counter()
a.add()
print(a.value, b.value)
```

Output: _____ Confidence: _____

Reason: _____

12 Halving

```
n = 8
steps = 0
while n > 1:
    n = n // 2
    steps += 1
print(steps)
```

Output: _____ Confidence: _____

Values of n after each division: _____

Heap work on paper

H1 Draw and update a heap

If heaps are included, draw the binary tree stored by this min-heap list: [2, 5, 3, 9, 7, 8, 4]. Put one value at each node. Otherwise, continue to question 13 on the next page.

Use the space below for your drawing. You may instead describe the levels and parent-child links in text.

Now remove the minimum. Show the last item moving to the root, then each sift-down swap. Use arrows on your drawing or write the intermediate lists.

Steps: _____

Returned value: _____ Final list: _____

Confidence: _____

Sorting costs and reasoning

13 Derive the running time

Fill in the tightest worst-case Big-O cost and give a short derivation for each sort. Use n for the number of items. Comparisons and moves take constant time. Assume distinct keys.

Use the taught versions: bubble sort may stop after a pass with no swaps; merge sort splits and merges; quick sort uses the last item as pivot and builds three lists. Write “not sure” when needed.

Algorithm	Worst case	Derivation or repeated work
Selection	_____	_____
Bubble	_____	_____
Insertion	_____	_____
Merge	_____	_____
Quick	_____	_____

Explain one sort in pseudocode

Choose one algorithm from the table. Write its steps in plain-language pseudocode, using indentation for loops or decisions. Do not write Python code. Then connect those steps to its worst-case cost.

Chosen algorithm: _____ Confidence: _____

An input pattern that reaches its worst case: _____

Why the steps above give that cost: _____
